

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer each and every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Additional Practice

1. Calculate the following indefinite integrals.

a) $\int x\sqrt{1+x^2} dx$

Let $u = 1+x^2$, then $du = 2xdx \rightarrow \frac{1}{2} du = xdx$

Therefore,

$$\int x\sqrt{1+x^2} dx = \int \frac{1}{2} \sqrt{u} du = \frac{1}{3} u^{3/2} + C = \frac{1}{3} (1+x^2)^{3/2} + C$$

b) $\int \frac{\sin(\sqrt{x})}{2\sqrt{x}} dx$

Let $u = \sqrt{x}$, then $du = \frac{1}{2\sqrt{x}} dx$

Therefore,

$$\int \frac{\sin(\sqrt{x})}{2\sqrt{x}} dx = \int \sin(u) du = -\cos(u) + C = -\cos(\sqrt{x}) + C$$

c) $\int \frac{2+\ln(x)}{x} dx$

Let $u = 2+\ln(x)$, then $du = \frac{1}{x} dx$

Therefore,

$$\int \frac{2+\ln(x)}{x} dx = \int u du = \frac{1}{2} u^2 + C = \frac{1}{2} (2+\ln(x))^2 + C$$

d) $\int \frac{2x}{1+x^4} dx$

Let $u = x^2$, then $du = 2x dx$

Therefore,

$$\int \frac{2x}{1+x^4} dx = \int \frac{2x}{1+(x^2)^2} dx = \int \frac{1}{1+u^2} du = \tan^{-1}(u) + C = \tan^{-1}(x^2) + C$$

e) $\int x^2 (1-x^3)^{10} dx$

Let $u = 1-x^3$, then $du = -3x^2 dx \rightarrow -\frac{1}{3} du = x^2 dx$

Therefore,

$$\int x^2 (1-x^3)^{10} dx = \int -\frac{1}{3} (u)^{10} du = -\frac{1}{33} u^{11} + C = -\frac{1}{33} (1-x^3)^{11} + C$$

2. Suppose that f is an odd function and g is an even function which are each integrable on the interval $[-3, 3]$. Given that $\int_0^3 f(x) dx = 4$ and $\int_0^3 g(x) dx = 5$, evaluate the following definite integral.

$$\int_{-3}^3 [6f(x) + 8g(x)] dx$$

We calculate that $\int_{-3}^3 [6f(x) + 8g(x)] dx = 6 \int_{-3}^3 f(x) dx + 8 \int_{-3}^3 g(x) dx$.

Since f is an odd function then $\int_{-3}^3 f(x) dx = 0$ and since g is an even function $\int_{-3}^3 g(x) dx = 2 \int_0^3 g(x) dx$.

Therefore, we have

$$\int_{-3}^3 [6f(x) + 8g(x)] dx = 6 \int_{-3}^3 f(x) dx + 8 \int_{-3}^3 g(x) dx = 6(0) + 8 \left(2 \int_0^3 g(x) dx \right) = 16(5) = 80$$

3. Calculate the following definite integrals.

a) $\int_0^{\pi/2} \sin(x) \cos(x) \sqrt{1 - \sin(x)} dx$

Let $u = 1 - \sin(x)$, then $du = -\cos(x) dx \rightarrow -du = \cos(x) dx$

Also note that $u = 1 - \sin(x) \rightarrow \sin(x) = 1 - u$

Our bounds become $u(0) = 1 - \sin(0) = 1$ & $u\left(\frac{\pi}{2}\right) = 1 - \sin\left(\frac{\pi}{2}\right) = 0$

Therefore,

$$\int_0^{\pi/2} \sin(x) \cos(x) \sqrt{1 - \sin(x)} dx = \int_1^0 -[1 - u] \sqrt{u} du = \int_1^0 [u^{3/2} - u^{1/2}] du = \left[\frac{2}{5} u^{5/2} - \frac{2}{3} u^{3/2} \right]_1^0 = [0] - \left[\frac{2}{5} - \frac{2}{3} \right] = \frac{4}{15}$$

b) $\int_0^1 x^3 \sqrt{3 + x^2} dx$

Let $u = 3 + x^2$, then $du = 2x dx \rightarrow \frac{1}{2} du = x dx$

Also note that $u = 3 + x^2 \rightarrow x^2 = u - 3$

Our bounds become $u(0) = 3 + (0)^2 = 3$ & $u(1) = 3 + (1)^2 = 4$

Therefore,

$$\begin{aligned} \int_0^1 x^3 \sqrt{3 + x^2} dx &= \int_3^4 \frac{1}{2} [u - 3] \sqrt{u} du = \int_3^4 \frac{1}{2} [u^{3/2} - 3u^{1/2}] du = \left[\frac{1}{5} u^{5/2} - u^{3/2} \right]_3^4 \\ &= \left[\frac{1}{5} (4)^{5/2} - (4)^{3/2} \right] - \left[\frac{1}{5} (3)^{5/2} - (3)^{3/2} \right] = \dots = \frac{6\sqrt{3} - 8}{5} \end{aligned}$$

c) $\int_0^{\pi/4} \cot(x) dx$

We see that $\int_0^{\pi/4} \cot(x) dx = \int_0^{\pi/4} \frac{\cos(x)}{\sin(x)} dx$. Clearly $\sin(x) = 0$ on $\left[0, \frac{\pi}{4}\right]$. So, $f(x) = \cot(x)$ is not

continuous on $\left[0, \frac{\pi}{4}\right]$. Therefore, we cannot continue.

d) $\int_{\pi/4}^{\pi/2} \cot(x) dx$

We see that $\int_{\pi/4}^{\pi/2} \cot(x) dx = \int_{\pi/4}^{\pi/2} \frac{\cos(x)}{\sin(x)} dx$

Let $u = \sin(x)$, then $du = \cos(x) dx$

Our bounds become $u\left(\frac{\pi}{4}\right) = \sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$ & $u\left(\frac{\pi}{2}\right) = \sin\left(\frac{\pi}{2}\right) = 1$

Therefore,

$$\int_{\pi/4}^{\pi/2} \cot(x) dx = \int_{\pi/4}^{\pi/2} \frac{\cos(x)}{\sin(x)} dx = \int_{\sqrt{2}/2}^1 \frac{1}{u} du = \left[\ln|u| \right]_{\sqrt{2}/2}^1 = \ln(1) - \ln\left(\frac{\sqrt{2}}{2}\right) = -\ln\left(\frac{\sqrt{2}}{2}\right)$$

e) $\int_0^1 x^2 \sqrt{1-x} dx$

Let $u = 1-x$, then $du = -dx \rightarrow -du = dx$

Also note that $u = 1-x \rightarrow x = 1-u$

Our bounds become $u(0) = 1-x = 1$ & $u(1) = 1-1 = 0$

Therefore,

$$\begin{aligned} \int_0^1 x^2 \sqrt{1-x} dx &= \int_1^0 -[1-u]^2 \sqrt{u} du = \int_1^0 [-u^2 + 2u - 1] u^{1/2} du = \int_1^0 [-u^{5/2} + 2u^{3/2} - u^{1/2}] du \\ &= \left[-\frac{2}{7} u^{7/2} + \frac{4}{5} u^{5/2} - \frac{2}{3} u^{3/2} \right]_1^0 = [0] - \left[-\frac{2}{7} + \frac{4}{5} - \frac{2}{3} \right] = \frac{16}{105} \end{aligned}$$

f) $\int_{-e}^e \frac{x^3}{\ln(x^2+2)} dx$

We cannot find a substitution will work here, but we notice that the integral is over the interval $[-e, e]$,

therefore we look to see if $f(x) = \frac{x^3}{\ln(x^2+2)}$ is either even or odd.

We see that $f(-x) = \frac{(-x)^3}{\ln((-x)^2+2)} = \frac{-x^3}{\ln(x^2+2)} = -\frac{x^3}{\ln(x^2+2)} = -f(x)$ which implies that $f(x)$ is an

odd function. Therefore, we know that $\int_{-e}^e \frac{x^3}{\ln(x^2+2)} dx = 0$.